

Forest Rail

The Jingzhang Railway Heritage Park already left the rails in the city. This proposal does not reinvent that park or turn it into a tech showcase.

Core claim: Forest Rail is the park-layer operating system along the Jingzhang heritage corridor—rusted rails for wayfinding, canopy for calm, Forest Rail terminal regulating seats / lights / daily goods. Zhongzhiyuan, Origin Community, and Dazhongsi differ in role but share one safety partition, human fallback, and exit rules [source:AGENT-TASKBOOK].

Forest Rail means:

- Forest: All three stations work as park layer—spaced trees, canopy, soft paving. Look out and see trees, not a glass showcase. The park quiets people; human–nature comes first.
- Rail (dual, separated): Heritage rails are the urban clue. Decorative guiding rails give direction; AI operating rails run safety-first shuttles for people + cargo, physically separated from walking.
- Forest Rail terminal: regulates public open seats / shared desks, low-glare ground edge lights, unbranded daily-essentials supply, and node open/close. No face ID, no brand ads.
- Three nodes: `FR-NODE-ZZY` / `FR-NODE-YY` / `FR-NODE-DZS` attach to the three key areas and can relocate when official polygons arrive [data:geometry/publicspace.geojson#FR-NODE-DZS] [metric:forestrailnodecount].

Refusals: district-wide building control, viral landmarks, light shows, planting strips posing as parks, private pods as “shared desks”, climbable decorative rails, smart-for-surveillance swaps, brand-store matrices.

Only one rail-guided corridor is added. If AI fails, the park body (soft paving, canopy, minimum seats, human service) still stands alone.

Executive Brief

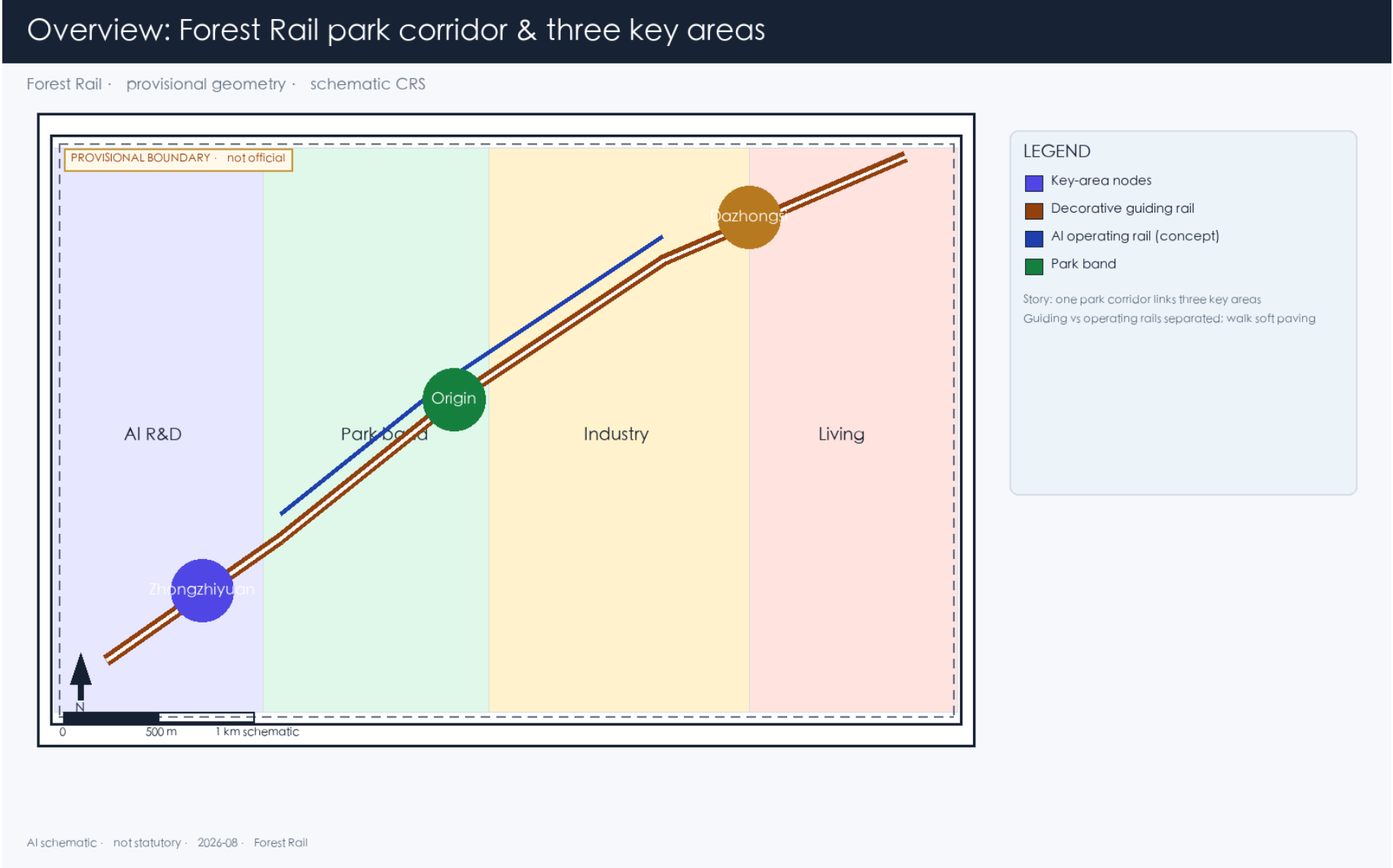
Design Basis and Source List

This proposal follows the Pre-qualification Announcement [source:OFFICIAL-ANNOUNCEMENT] and machine-readable site package materials [source:SITE-PACKAGE].

Current data uses provisional boundaries (`geometry/siteboundary.geojson`, `geometry/keyareas.geojson`) with `officialboundary=false`. Recalculate everything when official polygons arrive. This does not block content scoring; spatial conclusions remain discussion-grade.

Background and provisional sources are not elevated to statutory authority [source:SOURCE-REGISTRY].

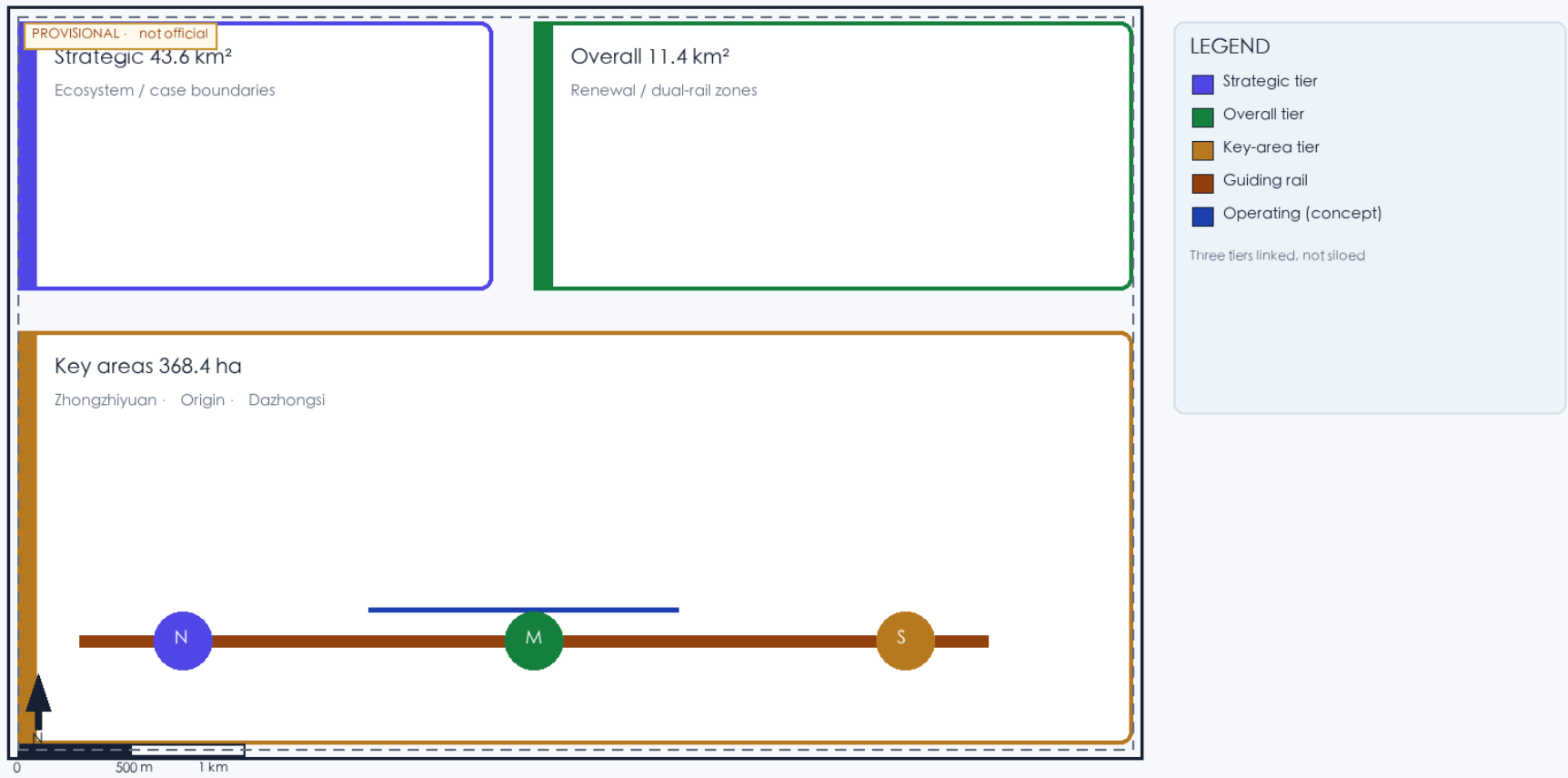
Authority order: GeoJSON → metrics → matrices → sources/assumptions/selfcheck → proposal → figures → HTML → PDF.



Three-Level Scope Framework

Three-tier scope and rail-guided structure

Forest Rail · provisional geometry · schematic



AI schematic · not statutory · 2026-08 · Forest Rail

Coordinated Research: Industry and Future City

Jingzhang is the existing narrative line—no new myth needed. 1909 opening; 2019 undergrounding left a surface heritage-park corridor [source:AGENT-TASKBOOK].

Narrow judgment: do not invent a parallel greenway; connect the heritage park into three-station daily life. Rust stays; people walk soft paving between rails. Zhongzhiyuan meets Qinghe; Origin meets campuses; Dazhongsi meets metro.

Naming: Forest Rail is both title and spatial description. International motif: Forest Rail — the park layer of an AI city. Minimal mark: rust twin rails + three canopy-green nodes in `assets/forest-rail-mark.svg` (see agent.5).

Urban / linear park cases (background only)

Park cases are background mechanism inspiration only; they do not replace the taskbook AI-ecosystem deliverable.

Sources: [source:CASE-CENTRAL-PARK]–[source:CASE-JZ-HERITAGE-PARK].

Urban life organism · international motif & minimal mark (agent.5)

Claim: Forest Rail — the park layer of an AI city.

Park/forest is a regulator—soft paving and canopy quiet the mind before creativity. Public carriers are shared desks and rest, not office lobbies outdoors.

Jingzhang three acts: (1) 1909 daring engineering; (2) 2019 rails left as park; (3) this proposal—three stations on the park layer; safety first; fail back to walking and humans.

Global AI innovation ecosystem cases & eight-factor map (agent.2)

Sources: [source:CASE-AI-STANFORD-RP]–[source:CASE-AI-TORONTO-VECTOR].

Urban agent protocol: walking-gap detection (auditable)

The urban agent does not replace planning approval and does not output personal profiles.

Industry test scenarios (agent.3 · ≥5 · six fields)

[metric:industrytestscenariocount] = 5. Tabletop: `node visual/assets/rundesktop.js --check` → [data:visual/assets/desk-tabletop-evidence.json]. Until G2 is assigned, limited trial is NOT AUTHORIZED [data:visual/assets/operations-matrix.json] [metric:deskpilotgatecount]. AV gates: [data:visual/assets/av-safety-gates.json].

Overall Design: Corridor Renewal and Park-Layer Systems

Corridor structure and ordinary complete path

Ordinary complete path (works without AI): static wayfinding → continuous accessible soft paving → shade/rest → staffed service → optional digital aid. Refusing AI must not lower basic service.

System grammar: continuous / withdrawable / fail-back

Lateral seams (relation interfaces)

Four systems

1) Shared Desk Protocol

Objects: under-canopy desks, bend seats, shelter hot desks [metric:deskzonecount]. AI: noise/rain/density → open/close + power. Hard rule: [metric:minopenseatscount]; no identity; no forced scan; wheelchair seats kept. Evidence: desk tabletop assets. G2 unassigned ⇒ NOT AUTHORIZED.

2) Safety-First AV Rail

Decorative ≠ operating; physical isolation; E-stop / breach / fail-stop; yield to pedestrians. Missing safety case/operator/drill ⇒ NOT AUTHORIZED. No unregistered precise km claims.

3) Edge Light Protocol

Soft-paving edges + seat zones only [metric:lightzonecount]. No lights on decorative rails, no chase/projection/show. Fault → off or ultra-dim.

4) Unbranded Daily Essentials

Trials [metric:dailytrialpointcount] → fit → open-admission stalls / quick-turnover shops. Fair siting keeps low-demand minimums; weekly bias audit; no face/ads.

Accessibility and inclusion

Wheelchair: prefer flat; if slope ≤1:20 + handrail. Deaf/blind: tactile + low-glare lights; concept dedicated AV shuttle (tactile/voice/text/help). Low-digital / non-Chinese: no forced app; bilingual icons. Fair siting + appeal + human override.

Safety section

AV collision/breach/E-stop fail → immediate stop, walk priority. Glare → dim/off. Climbing rust rails forbidden. No face ID. Offline → printed hours + human window.

Land Use, Building Scale, and Retain-Renovate-Demolish Strategy

Renewal within the overall design scope falls into three categories:

- Retain: Existing quality buildings along the heritage corridor — unchanged
- Renovate: Buildings needing ground-floor program and facade adjustment — remove LED screens, increase under-canopy transparency
- Renew: New construction on identified low-efficiency sites — building height controlled below tree canopy

Land use classification follows national standards [standard:MNR-LAND-USE-CLASSIFICATION-GUIDE], covering the design boundary without overlap [data:geometry/landuse.geojson#LU-001]. Building footprints and retain/renovate/renew classification are recorded in [data:geometry/buildings.geojson#BLDG-001].

Where FAR, building height, or road setbacks lack official data, they are marked as unknown. No numbers are fabricated.

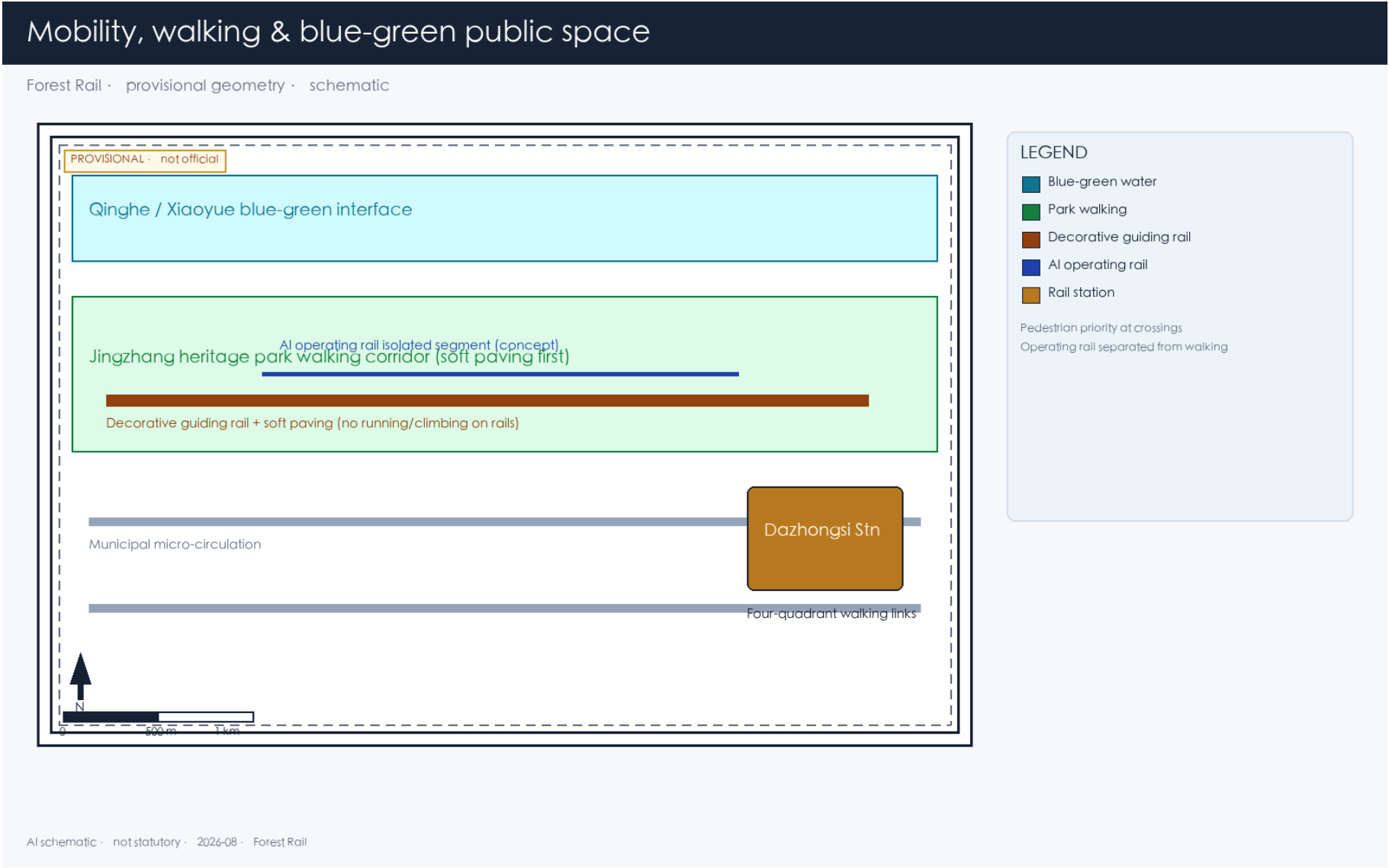
Transport, Rail, Municipal Infrastructure, and Public Services

Transport responds to the announcement's requirements for rail station integration, road micro-circulation, walking network gaps, and green transport [depth:trafficrailslowparking].

Key interventions:

1. Dazhongsi Station four-quadrant pedestrian connection: Metro Line 13 Dazhongsi Station — establish walking connections across all four quadrants of the intersection, with rail paths extending from station exits into the district
2. North Fifth Ring crossing node: Pedestrian linkage where the heritage corridor crosses the North Fifth Ring Road (underpass or overhead facility) to maintain north-south continuity
3. Rail path / municipal road intersections: Pedestrians have priority at crossings; rail tracks embedded in road surface indicate direction; no traffic signals govern the rail path

Road and walking network layers remain within the submission boundary [data:geometry/roads.geojson#ROAD-001], cross-checked with public space and green space. With a provisional boundary, transport conclusions serve as design discussion only.



Municipal infrastructure covers edge-computing nodes, distributed energy, and traditional utility integration. Missing utility, energy, and drainage data are listed as prerequisites for detailed design, not fabricated.

Blue-Green Network, Public Space, and Urban Character

Blue-green space follows the Jingzhang heritage corridor as its spine, coordinating Qinghe River, Xiaoyue River, and surrounding community movement [depth:bluegreenpublicspace].

The tree canopy layer is the primary character control mechanism: native trees with crown spread ≥8m (Chinese Scholar Tree, ash, ginkgo, goldenrain tree), targeting ≥70% coverage of building rooftops. Trees are not pruned into geometric shapes. Building height below the canopy — this is the basic criterion for the urban forest.

Character principles:

- Building facades use materials that age: brick, concrete, wood. No smooth glass curtain walls
- Rust and grey-green are the base colors, derived from oxidized rail and tree canopy
- Rooftops are for greening or equipment, not sculptural form
- Public art accepts only site-history-related, aging-compatible works — iron, stone, wood. No stainless steel or lighting installations

Which elements are design recommendations versus those pending official confirmation is itemized in `standardmatrix.json`.

Key Area Detailed Design

All three stations use the same structural block (location / park-layer role / Forest Rail terminal / AV/logistics / five components / must not / test-card binding), at equal depth—avoiding a Dazhongsi-only "showcase end". Current polygons are provisional [data:geometry/keyareas.geojson#PROV-KEY-001].

Zhongzhiyuan AI Autonomous Innovation Accelerator

Location: Northern end, along Qinghe River [data:geometry/keyareas.geojson#PROV-KEY-001]; node [data:geometry/publicspace.geojson#FR-NODE-ZZY].

Park-layer role: Qinghe waterfront quiet verification interface—set back buildings, under-canopy walks and stormwater grass slopes; safety evaluation translated into by-appointment collaboration nodes, not open attractions.

Forest Rail terminal: Appointment-only safety-node open/close; visitor seats retain minimum supply; riverside edge low-glare lights; optional unbranded daily points.

AV/logistics (concept): Isolated-segment material/visitor short-haul extension only; not mixed with pedestrian flow before authorization.

Must not: Open-spectator safety evaluations; riverside light-show facades; personal profiling; autonomous delivery competing with pedestrian time; claiming an evaluation venue already built.

Test-card binding: TEST-04 safety-node booking.

Beijing AI Origin Community

Location: Middle section, near universities [data:geometry/keyareas.geojson#PROV-KEY-002]; node [data:geometry/publicspace.geojson#FR-NODE-YY].

Park-layer role: Near-campus under-canopy co-learning / open-source third space—rail paths stitch campus gates to the park, breaking wall gaps.

Forest Rail terminal: 24h collaboration zones open/close by noise threshold; hot-desk power; quiet/discussion partitions; minimum seats and human service window.

AV/logistics (concept): Campus–park restock and accessible shuttle concept; human fallback before maturity.

Must not: talent-apartment-complex stacking; light shows; developer profiling; private-pod desks; claiming release hall already formally open.

Test-card binding: TEST-05 near-campus desk fairness.

Dazhongsi AI Industry Cluster — Urban Forest

Location: Southern end, adjacent to Metro Line 13 Dazhongsi Station [data:geometry/keyareas.geojson#PROV-KEY-003]; node [data:geometry/publicspace.geojson#FR-NODE-DZS].

Park-layer role: Station-into-park + urban-forest daily life—exit the station into a park; find a shared desk in that park.

Forest Rail terminal: Full seats/lights/daily trial operation; offline revert to printed rules.

AV/logistics (concept): Daily restock + deaf/blind accessible shuttle extension concept; partitioned from walking; not commercial claims.

Must not: brand-store matrix; face monitoring/ad push; light shows; extinguishing all seats; claiming AV already commercial.

Dazhongsi Station integration: From exits A/B/C/D, a rail path enters each quadrant. First view is park and rusted rail—not ad screens.

Commercial services: Unbranded daily trial points → fit scoring → open-admission stalls / community quick-turnover shops; low-demand nodes keep minimum supply.

Test-card binding: TEST-01 / TEST-02 / TEST-03.

Concept views (not formal evidence figures; spatial claims remain in GeoJSON / metrics / the five evidence diagrams):







Three key areas — task index

Forest Rail · provisional geometry · schematic

Zhongzhiyuan (N)

- Qinghe edge
- AI safety nodes
- By appointment

轨径 / rail path

Origin Community (M)

- Campus transfer
- Open-source collab
- Walking stitches

轨径 / rail path

Dazhongsi (S)

- Exit → park
- Shared desks
- Dual-rail complete

轨径 / rail path

LEGEND

- N key area
- M key area
- S key area
- Rail path

PROV-KEY-001/002/003 not official



AI schematic · not statutory · 2026-08 · Forest Rail

All three key areas appear in `geometry/keyareas.geojson`. Current data is provisional constraint; the proposal, HTML, and selfcheck state that it cannot serve as approval or scoring basis.

A Day Here: Personas and Scenarios

Five User Personas

Scenario Cards (≥10 · six-field protocol)

Each card uses six fields: service audience / minimum necessary data / spatial boundary / human reviewer / exit/appeal / stop condition. Table below is summary; full fields expand in test cards and operations matrix. [metric:scenariocardcount]

All AI governance follows data minimization, open sourcing, explainability, and human review. Seat/daily-point open/close must retain staffed help and minimum public supply; siting algorithms must not chase high footfall at the expense of basic service in low-demand communities. Urban intelligent agents assist in identifying walking network gaps, public space activity, and facility maintenance needs but do not replace planning approvals or output unauthorized personal profiles.

Component Library and Quiet Honors (agent.4 · lightweight)

No viral check-in leaderboard. Reusable components and "quiet honors" record:

Three quiet landmarks (not attractions) remain: rusted rail curve with old scholar tree; rail path–Xiaoyue River crushed-stone square; old platform retaining wall.

Annual Activities and Long-Term Operations (agent.6 · mechanism)

International outreach uses walkable visit routes and public evidence packs only—no parade stages.

Renewal Project List and Phasing

Near-term pilots can start with lightweight interventions (spread crushed stone, place benches, plant saplings) without waiting for all engineering conditions. Medium and long-term projects require confirmed regulatory plans, municipal engineering, and property rights.

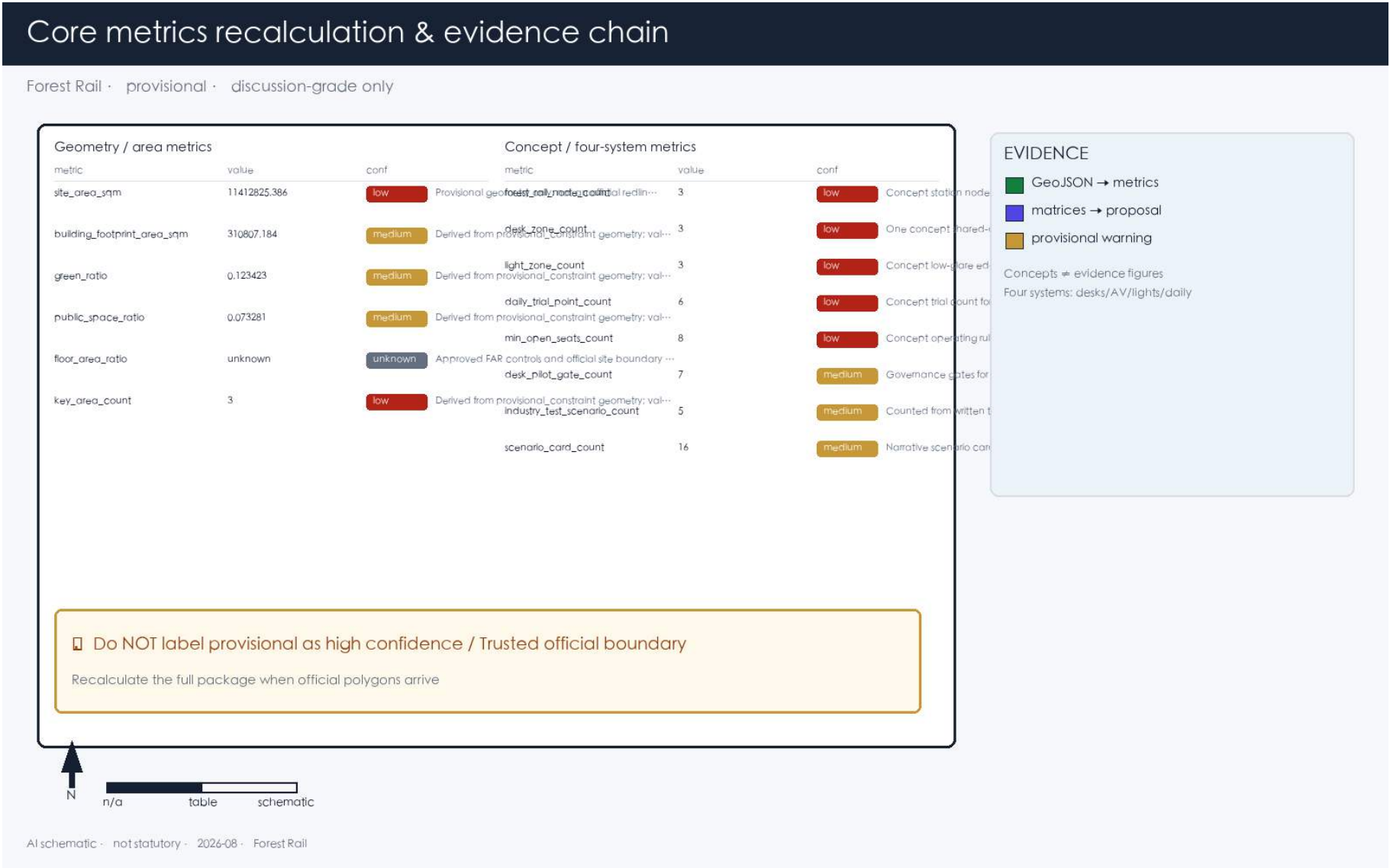
Prerequisites checklist (actors, rights, cost band, acceptance, ops): see `report/narrative.md` [source:NARRATIVE-SUPPORT].

Phasing spatial evidence [data:geometry/phasing.geojson#PHASE-001].

Metrics Framework

Metrics are divided into three categories:

Three categories enter `metrics.json`, `assumptions.json`, and `compliancematrix.json` respectively. Full calculations are in `metrics.json`; this text explains design intent only.



- Bilingual check + prerequisites: `report/narrative.md` [source:NARRATIVE-SUPPORT]
- Figure footers cite package geometry/metrics + 2026-08 only — no hallucinated agencies, XX-city placeholders, or Working Draft seals

Missing Data

Gaps listed in `missingdatachecklist.csv` — official boundaries, regulatory plans, roads, municipal engineering, heritage protection — enter `assumptions.json` and self-check. Conclusions lacking official conditions are downgraded to pending confirmation.

Copyright (display)

The proposal provides bilingual versions (this file and `proposal.md`). Images, data, and code assets are documented in `sources.json` and `report/copyrightstatement.md`. HTML pages do not load remote scripts, map tiles, fonts, iframes, or external APIs.

Disclaimer

This proposal does not claim official approval, regulatory plan status, final property rights, final construction scale, or implementation guarantee.

References

See [source:OFFICIAL-ANNOUNCEMENT], [source:AGENT-TASKBOOK], [source:SITE-PACKAGE], `sources.json`, `metrics.json`, `compliancematrix.json`.

- brief/public-brief.md
- brief/site-package/designbrief.json
- brief/site-package/alloweddesignspace.json
- data/processed/agentfactpack.md
- data/processed/projectscopesummary.csv
- data/processed/agenttaskrequirements.csv
- data/processed/missingdatachecklist.csv
- Full source index: `sources.json`, `metrics.json`, `compliancematrix.json`, `standardmatrix.json`, `designdepthmatrix.json`